Juanita Cathy

6X Hackathon Winner - Co-Founder NovaCopy AI - Public Speaker

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EDUCATION

Anna University *Chennai, IN*
B.E Computer Science and Engineering ***Aug 2022-Aug 2026***
GPA: 9.27/10
3x Consecutive Rank 1 · President of Hacker Society · MLSA · Google Developer Groups On Campus Lead

St John's English School and Junior College *Chennai, IN*
HSC (class 12) : **84.5%** · **SSC** (class 10): **98%** [Rank 2] ***Apr 2010-2022***

SCHOLASTIC ACHIEVEMENTS

- Achieved ***Rank 2*** in ***CBSE Class 10 Board Exams*** and was awarded for academic excellence in a competitive environment
- Awarded ***Rank 1 Holder for 3 consecutive years (1-3,5 sems)*** in **Anna University Exams** at my institution and ***Rank 2*** in 4th sem
- Secured ***3rd Rank*** institution-rank in ***Science Olympiad*** and was awarded with monetary prize of Rs.2000

SKILLS

Programming Languages Proficient – Python · JavaScript · TypeScript · Java | Intermediate –· C · PHP

AI & Machine Learning TensorFlow · PyTorch · Scikit-learn · Hugging Face Transformers · OpenCV · Langchain

Web & Software Development React.js · Next.js · Node.js · Express.js · HTML/CSS · Firebase · PostgreSQL · Drizzle ORM

DevOps Docker · Kubernetes · GCP · Azure **Tools** Git · Jupyter Notebook · WebSockets (Socket.io) · getStreamio

RESEARCH EXPERIENCE

Advanced Deep Learning Approach for Palm-Leaf Manuscript Character Recognition [Preprint] *Chennai, India*
Research Intern | Mentor: Dr.S.Sivagami | Jeppiaar University ***Dec 2024 - Feb 2025***

- Developed a Tamil character recognition system for ancient manuscripts, achieving an accuracy of 92.38% on manuscripts from Mendeley.
- Trained a ResNet-50 model on the HPL-Tamil dataset, attaining a high accuracy of 96%
- Preprocessing techniques such as CLAHE, adaptive thresholding, and character segmentation using the Watershed algorithms were applied to refine recognition.

Enhancing Emotional Intelligence in LLM Responses Using Speech Emotion Recognition [Preprint]/ [GHUpload] *Chennai, India*
Independent Research | Mentor: Dr. Arockia Renjith ***Sept 2024 - Dec 2024***

- Designed a hybrid deep learning approach for SER integrating CNN and BiLSTM networks for emotion classification.
- Processed speech inputs from four benchmark datasets**,** training on 12,162 labeled samples achieving a **99.7%** training accuracy and **84.7%** validation accuracy.
- Outperformed SVM (78.57%) and Random Forest (71%) in accuracy demonstrating the effectiveness of deep learning-based SER for enhancing emotionally intelligent HCI.

WORK EXPERIENCE

C1X Inc. *Chennai*
SOFTWARE ENGINEERING INTERN ***July 2025 - Oct 2025***

- Architected a **multi-agent conversation system** using Server-Sent Events (SSE) and Gemini, enabling real-time orchestration between LLM agents for campaign form generation, improved response latency and adaptability by ~45%
- Developed an intelligent state management and tool-calling pipeline leveraging LLM-driven reasoning and agent collaboration, achieving 35% faster data enrichment and enhancing contextual accuracy.
- Engineered an end-to-end OCR + LLM Request for Proposal (RFPs) processing system that automatically extracted and structured RFP information from varied email formats and image files.
- Collaborated in defining testing strategies using LLM-as-Judge and LLM Test Mate (AWS) for evaluating LLM correctness and consistency

HEADSTARTER AI *Remote*
SOFTWARE ENGINEERING FELLOW ***Jul 2024 - Dec 2024***

- Built 5+ AI apps (link) and APIs using NextJS, OpenAI, Pinecone, StripeAPI etc within a given deadline
- Created and Shipped an AI Product within the fellowship and gained traction from over 5k users
- Developed large scale architecture projects from design to deployment using MVC design patterns with a team
- Coached by Amazon, Bloomberg, Capital One engineers on Agile, CI/CD, Git and microservices with Leetcode mocks

PROJECTS

NovaCopy AI (150+ waitlists, 5k visits in 24 hrs) *(~150+ hours)* | 🔗 ***Aug 2024- Present***

- Built a **full-stack AI-powered copywriting assistant** that helps users generate high-quality marketing and creative content
- Achieved **150+ waitlist signups** and over **5,000 site visits 24 hrs of the launch of the project.** Designed, developed and deployed website using Next.js, TypeScript, Vercel, Stripe.
- Integrated **customizable templates, tone control, and SEO optimization features to improve content relevance**

DevOrbit – Connect Random Devs around the world! *(Most Creative Hack Winner-BoostHacks)* 🔗 ***June 2024***

- Leveraged **Next.js, TypeScript, Tailwind CSS, WebSockets, and PostgreSQL** to create a platform to pair program, share code snips and works in rooms (public or private) and has language based tagging to filter the rooms. Supports 100+ rooms.

MumCares - Postpartum Mental Health Analysis with AI *(SheHacks Top 10)* 🔗 ***April 2024***

- *PROBLEM SOLVED* : Postpartum Depression of new mothers by analyzing sentiment and emotional cues.
- Developed using React, Python, Flask and concepts of **NLP models, sentiment analysis, and SER**

Forger.AI *(~24 hours) |* ***(Best Real-world Hack)*** 🔗 ***Sept 2024***

- Utilized TypeScript and Python to create an application to personalized cold emails based on job postings and send them to potential recruiters. Uses Langchain for managing models, Groqcloud, FastAPI and SendGrid API.

RELEVANT COURSEWORK

Core Computer Science: Data Structures and Algorithms (DSA) · Object-Oriented Programming · DBMS · System Design · Operating Systems · Computer Networks · Distributed Systems · TCP/IP Networking Protocols · Linux
Artificial Intelligence & Machine Learning (HONOURS): Neural Networks and Deep Learning · Soft Computing · Optimization Techniques · Fundamentals of Data Science · Natural Language Processing
Mathematics & Statistics: Linear Algebra · Probability and Statistics · Discrete Mathematics ·

RELEVANT CERTIFICATIONS

- ***CCNA CISCO CERTIFIED IN NETWORKING (Graded)*** 🔗
- ***EVERYTHING ABOUT DEEP LEARNING, SkillArena (Graded, 9.2/10)***
- ***NPTEL DATA STRUCTURES AND ALGORITHMS, IIT MADRAS (Graded, Qualified)***
- ***HCLTech: Artificial Intelligence Project Based Learning*** *(Naan Mudhalvan)* ***(Graded, 10/10)***
- ***AZURE AI DOCUMENT INTELLIGENCE CUSTOM SKILL FOR COGNITIVE SEARCH,*** *Microsoft Certification* 🔗
- ***TEXT IN IMAGES AND DOCUMENTS USING AI,*** *Microsoft Certification* 🔗
- *Microsoft Course Achievements Profile*: Link 🔗

HACKATHON ACHIEVEMENTS

- Won ***Most Creative Hack Award*** at ***BoostHacks 2024*** for the project DevOrbit, a developer platform enabling real-time code collaboration. 🔗
- Got the award for ***Best Use of Taipy*** at ***MLH All in Open Source Hackathon*** for project CO2Vision, a carbon emissions tracking project. 🔗
- Won ***Best Finance Project*** at ***Nosu Hacks*** for Purrse, a cat themed financial tracker application for broke students.
- **Top 10 Finalists** at SheHacks by IIIT for the project MumCares AI. 🔗
- Secured 4th Place in ***Seam Mini App Challenge*** for project Haikura, sakura themed mini app.

POSITIONS OF RESPONSIBILITIES

- ***Google Developer Groups On Campus Lead*:** Organized workshops and developer meetups to promote open-source and Google technologies.
- ***Microsoft Learn Student Ambassador:*** Conducted AI/ML and cloud-related sessions for students.
- ***Coordinator for Innovision Presentation Event:*** Managed event planning, logistics, and speaker coordination.
- ***Head of English Club:*** Conducted debates, writing contests, and public speaking sessions.

AWARDS

University Rank 1 Award (3 Consecutive Sems)
Awarded for achieving rank 1 in university exams.

Best Speaker and Technical Presenter
Awarded in the JEC Awards Ceremony

6x Hackathon Winner
2k+ in prizes from attending 10+ hackathons

UF Quantum Computing OPEN SOURCE Winner
One of 66 winners from 800+ participants across the globe.